Evidence-Monotone Audio Representation: Preventing Provenance Promotion Across Derived Audio Transformations

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Abstract

Derived audio can keep its exact waveform while silently strengthening what it says about its own origin. Under a minimal boundary-only inheritance policy (a constructed reference policy, not the behaviour of any named product), a clip assembled from captured and generated material, then trimmed, resampled and re-encoded, emerges described as captured-only. We specify an operator contract for derived audio: provenance is the union over the complete required source set, same-channel support is the minimum over it, applicability is intersected, lineage is the union, and an unverified source interval is absorbing. The underlying algebra is an instance of a known meet-semilattice construction and is not claimed as new. What is new is the executable temporal instantiation over sample-exact intervals with declared kernel footprints, a signed transport that carries interval evidence inside C2PA temporal regions of interest, and a measured demonstration. No model is trained and no acoustic feature is used. Across 885 828 contract checks, 10 000 adversarial timelines, 600 mixed-origin clips transformed by stock FFmpeg, and 10 860 cases of a two-language differential oracle, complete-source propagation produced 0 contract failures, 0 promotions and 0 lineage omissions, while boundary-only inheritance promoted on 94.5% of timelines and on 3 600 of 4 800 transformation runs. Decoded essence was identical before and after signing in 120 signed round-trips, with 0 mismatches, at a bookkeeping cost of 0.93 ms per audio-minute. The evidential cost of the footprints is measured rather than assumed: dilution is confined to fixed-radius bands at ancestry boundaries, and for the fixed edit chain and boundary geometry tested it falls approximately in inverse proportion to asset duration. The contract constrains what a derived representation may claim about its sources; it does not detect deepfakes, does not establish that speech is truthful, and a cryptographically valid signature proves only that an assertion was signed, not that it is true.

Keywords: digital evidence, media provenance, content credentials, C2PA, audio forensics, synthetic speech, chain of custody

1. Introduction

Speech can now be generated convincingly enough that the question “is this recording real?” is answered less and less reliably by listening to it. Two different mechanisms have grown up in response. Detection asks whether a signal carries the traces of synthesis [1, 2]. Provenance asks what

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an asset's own cryptographically signed history says about where it came from [3]. These answer different questions, and this work is about the second one, so nothing here is a detection result.

Provenance systems bind assertions to an asset. They do not, on their own, say what a derived asset is entitled to assert. That gap opens as soon as audio is edited, because editing is routine: a broadcast clip is trimmed, a podcast segment is spliced, a submitted exhibit is resampled and re-encoded before it reaches an examiner. Each of those steps produces a new asset whose manifest is written by whatever software performed the step. The C2PA specification describes the relationship such an asset has to its input with the ingredient relationship `parentOf`, whose meaning is given as "The current asset is a derived asset or asset rendition of this ingredient" [3, Table 10, §18.16.3]. The specification's vocabulary is in fact rich enough to carry everything the problem needs: a composed asset records each component as a `componentOf` ingredient with its own manifest history, a generative origin can be marked with a `digitalSourceType`, and an assertion can be scoped to a temporal range. What the text does not prescribe, and we searched it for such a rule, is the *reduction*: the aggregate evidence state a derived interval should emit when the component histories it represents disagree. The specification can preserve the component histories; it does not define the conservative claim that must be computed over them. The coalition's July 2026 implementation guidance on identifying synthetic and non-synthetic content sharpens the same boundary from the other side [4]: it develops the vocabulary for declaring AI involvement, including `digitalSourceType`, an AI-disclosure assertion and regions of interest that localise which segments of audio or video were affected. That is guidance on how to *represent* what happened to part of an asset, and it is complementary to what this paper supplies, which is the rule for what a derived interval may *claim* once its parts disagree. That distinction is the entire subject of this paper, and it means the contribution here is not a competing provenance system but a semantic aggregation contract that operates inside the standard's own vocabulary.

The consequence is concrete. Consider five consecutive source intervals of a recording whose third interval is generated speech and whose other four are captured. A derived asset representing all five can be described from its retained boundaries, which are captured, and emerge labelled captured-only. The decoded audio is bit-identical either way. Only the evidential claim differs (Figure 1). In a forensic setting that difference is the whole point: an examiner reading a captured-only claim over an interval that contains generated speech is being told something stronger than the sources support, by a chain in which no signature was forged and no cryptographic primitive failed.

We call this *provenance promotion*: a derived representation acquiring a stronger evidential claim than the complete source material it represents, without any change to the decoded signal. The requirement that prevents it is what we call *evidence monotonicity*: along a chain of representation-only transformations, the emitted claim must be no stronger than the meet of every required source, ancestry accumulates rather than disappears, and the absence of verified evidence is absorbing.

An earlier instantiation of that requirement exists for spatial data, where derived terrain geometry can overstate the evidence of the source arc it replaces [5]. This work asks whether the requirement is usable in a temporal medium with a real signed transport and real processing software, and what it costs. It is not a re-presentation of that work: Table 1 states exactly what is inherited and what is new, no earlier result is reused, and the row-by-row transfer analysis is Supplementary Note S1. The short version is that the algebra transfers and almost nothing else does: temporal interval maps, the kernel-footprint problem and its monotonicity resolution, the guard-band validation against third-party processing, signal transparency on decoded PCM, and the signed transport have no spatial analogue.

Table 1: Relation to the spatial predecessor [5]. Inherited elements are prior work and are cited, not claimed; every reported number in this paper is new.

Inherited from the spatial work	New in this work
Evidence record (P, S, A, L); provenance as an atom set with mixed by union	Temporal required-source sets over half-open sample intervals
Same-channel meet by minimum; scope intersection; lineage union	The captured/generated alphabet and the capture-support channel
Absorbing unverified state	Requirement (iv): a channel not applicable to *every* required source is withheld, not computed from the rest
Requirement (iv′): a $\perp$ source also withholds every numeric channel	
Non-promotion in the spatial setting	Kernel footprints with declared source-support bounds, validated per clip against stock FFmpeg; footprint monotonicity (Proposition 4)
Conformance-test methodology (exhaustive enumeration, frozen fixtures, separately implemented oracle)	Signal transparency on decoded PCM; signed C2PA transport with temporal regions; the claim-dilution cost metric

We are equally explicit about what is *not* new. The algebra we use, a meet over a semilattice of labels with union of ancestry and an absorbing bottom element, is an instance of constructions that are decades old, in semiring-based provenance annotation [6] and in label lattices for secure information flow [7, 8]. We therefore do not claim the non-amplification property as a new theorem. We state it as an instance, prove the parts that are specific to the temporal setting, and locate the contribution in what is executable and measured.

Contributions. Each is traceable to a completed experiment.

1. An operator contract for derived media stated without reference to any modality, with its non-amplification property presented as an instance of a known meet-semilattice construction rather than as a new theorem (Section 4).
2. A temporal audio instantiation over sample-exact half-open intervals for eight representation operators, each with a declared conservative kernel footprint, together with a footprint-monotonicity result showing that over-approximating the required source set is always safe (Section 5, Proposition 4).
3. A signed transport in which interval-level evidence is carried inside C2PA temporal regions of interest and survives derivation and re-signing, demonstrated end-to-end on 3 audio containers with 120 round-trips, plus a C2PA-native heterogeneous composition in which signed captured and generated sources become `componentOf` ingredients with `c2pa.placed` references, under a documented profile reading of the action vocabulary, and the EM aggregate rides alongside (Sections 5.4, 7.5, 7.7).
4. A policy ablation showing that complete-source aggregation and declared kernel footprints are each necessary and neither sufficient: removing either one produces promotion on every timeline of the arm it fails (Section 7.3).

5. An executable conformance suite: 885 828 checks over every source word of length up to 8 over $\{C, G, \perp\}$, with 0 failures, and a two-language differential over 10 860 frozen cases with 0 disagreements (Sections 7.1, 7.9).
6. A measured demonstration on 600 mixed-origin clips processed entirely by stock FFmpeg, in which the interval model is validated against the behaviour of software we do not control and the declared footprints are shown adequate (Section 7.4).
7. A cost measurement on both axes that matter: runtime, where bookkeeping is 0.156% of the processing time it accompanies and the assertion grows linearly in evidence intervals at 333 B per interval (Section 7.10); and *evidential* cost, a claim-dilution metric showing that the conservatism the footprints buy is priced in samples per ancestry boundary, not in fractions of the asset (Section 7.8).

Three framing sentences apply throughout. The contract does not infer authenticity from acoustic features; it prevents a provenance-aware processing chain from promoting a mixed or unverified source interval into a stronger captured-only class. An unsigned or provenance-stripped asset is classified as unverified, never as synthetic. The contract is orthogonal to passive deepfake detectors and to watermarking systems, and composes with either.

Transparency obligations for synthetic content under Article 50 of Regulation (EU) 2024/1689 became applicable on 2 August 2026 [9], which makes machine-readable origin claims about derived media operationally relevant. We make no legal claim about what any particular disclosure regime requires.

2. Related work

Detection of synthetic speech. Anti-spoofing and deepfake-speech detection classify a signal by its acoustic traces, with challenge resources and surveys documenting both the progress and the generalisation gap when detectors meet unseen attacks or unseen codecs [2, 1]. What this line guarantees is a decision about a signal. What it does not do is constrain what a derived asset may assert about its own sources, and it needs no source evidence to exist. The two are complementary: a detector can be run on an asset whose evidence chain is intact, and the contract here says nothing about what the detector should conclude.

Watermarking and soft bindings. Localised audio watermarking marks generated speech at synthesis time and detects the mark per region [10]; C2PA treats watermarking and fingerprinting as *soft bindings* used to recover a manifest that has been separated from its asset [3, §18.10]. These guarantee recoverability and, in the localised case, region-level presence of a mark. They do not define what the aggregate evidence claim of a derived asset should be when its sources differ, and a region that carries no mark is not thereby captured.

Cryptographic media provenance. C2PA binds signed assertions and transformation history to an asset, defines ingredient relationships `parentOf`, `componentOf` and `inputTo`, and supports assertions scoped to part of an asset through regions of interest, including temporal ranges in normal play time [3, §18.2.2.3, §18.16.3]. Security analyses of the standard and its implementations report problems in the cryptographic and trust layers: timestamp disagreement, certificate revocation handling, and inconsistent validator behaviour [11]. Related work constructs assets whose manifest and whose watermark make contradictory claims that each pass their own verification, through ordinary editing

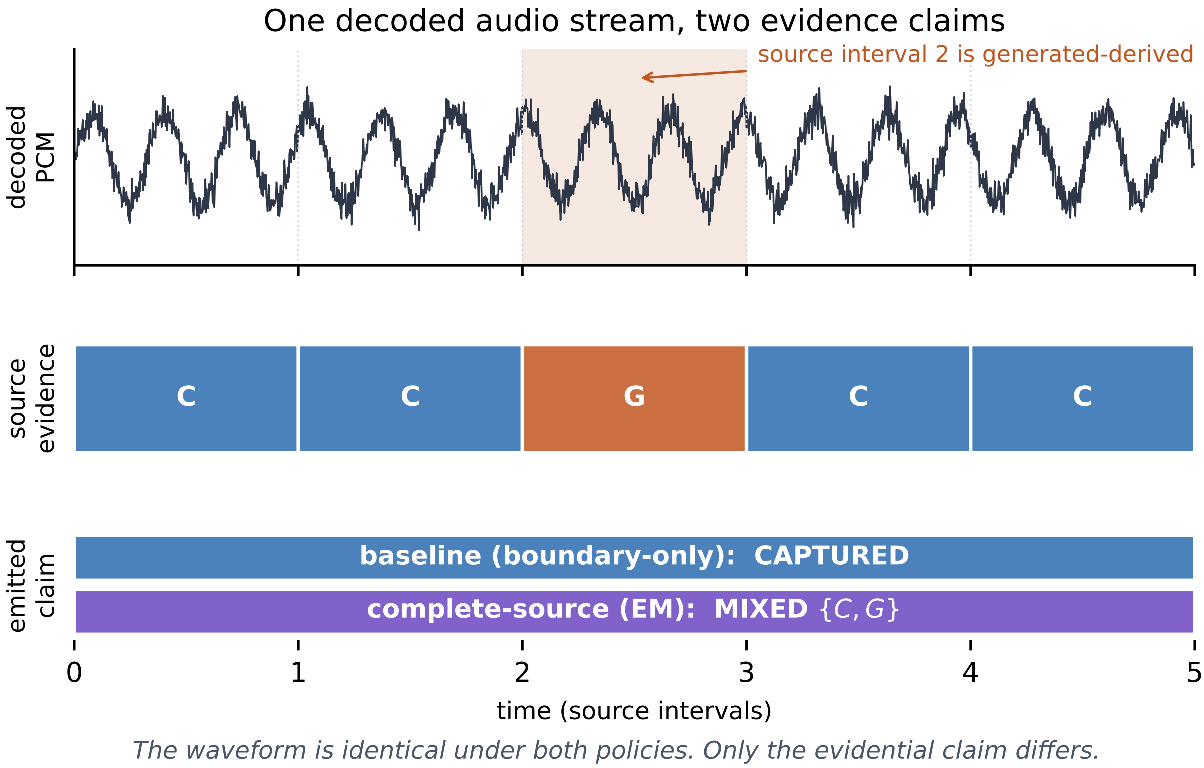

Figure 1: The minimal temporal counterexample. Five source intervals, of which the third is generated-derived. A boundary-only reading of the retained region sees captured evidence at both ends and may emit CAPTURED; the complete-source rule sees $C + C + G + C + C$ and emits MIXED $\{C, G\}$. The decoded audio is identical under both policies.

workflows and without any cryptographic compromise [12]. Our concern is a different layer again, and it is orthogonal to both: assuming keys are intact, signatures verify, and no assertion is omitted in bad faith, is the assertion that a conforming derived asset emits conservative with respect to every source interval it represents? The distinction matters because the specification already preserves a great deal: a `componentOf` ingredient carries its own manifest, composition is required to record the placed components, and a generative component can carry a `digitalSourceType` [3, §18.16.3]. A consumer walking the full ingredient graph could in principle reconstruct everything. What the text does not define is the aggregate: when a five-second derived interval represents four captured components and one generated one, no normative rule states that the interval's own evidence claim must be the conservative reduction over all five rather than an inheritance from its parent. We read the specification for a rule of that kind and did not find one; the specification governs how assertions are bound and validated, not how an aggregate evidence claim over heterogeneous sources must be computed. The reference implementation behaves accordingly, and our own composition experiment is the evidence: every one of those compositions carries two ingredients whose declared source types disagree, and the official validator reports each as valid without computing, requiring, or remarking on any aggregate claim over them (Section 7.7). The aggregate in those manifests is present because the contract put it there.

Inferring audio provenance from the signal. A separate line carries the name of this problem and solves a different half of it. Recent work formulates *audio provenance analysis* as the reconstruction of content reuse and parent-child relations across a heterogeneous collection with no prior knowledge of its contents [13], combining partial audio matching with phylogeny to build directed acyclic graphs that identify the least altered version in a near-duplicate cluster and trace reuse within it, including multi-source compositions and transcoded material. What that line guarantees is a reconstructed lineage where none was recorded. It is complementary to the problem here in the most direct way: their method infers a graph from an unknown signal, and the contract here assumes signed source evidence already exists and constrains how a known representation operator may aggregate and propagate it. Neither substitutes for the other, and a pipeline could reasonably use inference to recover a lineage and this contract to decide what the recovered lineage entitles a derived interval to claim.

Provenance for edited audio. Two further lines are the closest work that assumes evidence rather than inferring it. Chunk-localised speech provenance binds short time segments to an issuer-signed original through perceptual fingerprints and Merkle commitments, addressing exactly the fragility of hard bindings under re-encoding [14]. Zero-knowledge constructions prove that an edit was drawn from an allowed operation set while hiding the source, the edits and the device [15]. Both provide something this work does not: the first re-establishes the link between a time segment and a signed original after transport, the second proves edit correctness without disclosure. Neither states an aggregation rule for what the derived asset's evidence claim may be when the sources it represents disagree, and neither proves a non-amplification property.

Provenance models and information flow. W3C PROV records derivation as a graph [16]: a PROV record can state truthfully that a segment was derived from a source while a consumer still reads the segment's evidence from its boundaries. Concretely, a PROV graph for the five-interval counterexample would contain `wasDerivedFrom` edges to all five source intervals, a complete and true derivation record, and still say nothing about which *claim* the derived segment may emit, because PROV constrains the shape of the history, not the aggregation of evidential authority over it. The algebraic machinery for constraining that reading is long established. Provenance semirings annotate derived data with elements of a semiring, so that combination is associative and distributes over alternative derivations [6]. Idempotence is not part of that general framework and is in fact incompatible with several of its standard instances; the meet used here is idempotent, but that is a property of this particular structure rather than of semiring provenance at large. Lattice models of secure information flow classify objects and require that combination move only in the conservative direction [7, 8]. The property we need, that a combination is never stronger than any of its inputs, is the meet in such a structure, and we present it as such rather than as a discovery.

The gap. Table 2 places these lines side by side. Each row states what that line does guarantee before naming the property it leaves unenforced on a derived audio interval. No row provides an operator-level complete-source rule instantiated over media time and carried in a signed transport; that combination is what this work contributes, and the matrix behind the table, with search dates and query strings, is Supplementary Note S2.

Table 2: Positioning. Each row names a guarantee the line already provides, then the evidence property it does not, by itself, enforce on a derived audio interval. The contribution here is the joint enforcement of the right-hand column through representation-only operators, not the reinvention of any left-hand guarantee.

Line of work	Guarantee it provides	Left unenforced on a derived interval
Synthetic-speech detection [2, 1]	A decision about a signal from its acoustic traces	What the derived asset may assert about its own sources
Localised watermarking and soft bindings [10, 3]	Region-level presence of a mark; recovery of a separated manifest	Aggregation of heterogeneous source evidence into one claim
Cryptographic provenance [3]	Signed assertions, ingredient relationships, temporal regions of interest	A rule requiring the derived claim to be conservative over all ingredients
C2PA security analysis and cross-layer attacks [11, 12]	Weaknesses in the trust, timestamp and validator layers; contradictions between layers	Behaviour of a chain in which every layer is intact and honest
Audio provenance analysis [13]	Reconstruction of reuse and parent-child relations from unknown signals, via partial matching and phylogeny	What a derived interval may claim once its recorded evidence disagrees
Provenance for edited audio [14, 15]	Chunk-to-original binding after transport; zero-knowledge edit correctness	An aggregation rule when represented sources disagree
Provenance graphs and lineage [16]	A truthful record of derivation	Operator-level monotonicity as an executable contract
Provenance semirings and information-flow lattices [6, 7]	The algebra of conservative combination	A temporal media instantiation with a signed transport

3. Threat model

In scope. Provenance promotion and laundering through representation-only transformations, under four conditions: source evidence exists; the processing system holds the source-to-output mapping; the operator is expected to conform to the contract; and cryptographic primitives and signing keys are intact.

Out of scope. Capture devices that lie about what they recorded; compromised signing keys; dishonest certificate authorities; unsigned media of unknown origin; speaker identity; the truthfulness of what is said; semantic misinformation; waveform-artefact deepfake detection; watermark removal; and full provenance stripping followed by construction of a fresh, dishonest trust chain. Attacks on the cryptographic and trust layers are the subject of separate analyses [11] and are assumed away here precisely so that the residual semantic question is isolated.

Required behaviour when evidence is absent or invalid. The state is UNVERIFIED or INVALID. It is never automatically "fake" and never automatically "captured". An asset can lose its manifest for entirely benign reasons, and nothing in this work licenses reading a missing manifest as evidence of intent.

Malicious signer. Cryptographic validity proves that an assertion was signed, not that it is true. A signer willing to assert falsehoods can produce a valid manifest containing them, and no propagation rule inside the processing chain can prevent that. The contract constrains conforming operators; it is not a defence against a dishonest issuer. A threat-model matrix pairing attacker capability against contract behaviour and against what is explicitly not prevented is Supplementary Note S3.

4. Formal model

4.1. Evidence records

Let a source asset be partitioned into elements. Each element x carries an evidence record

$$e_x = (P_x, S_x, A_x, L_x), \tag{1}$$

where P_x is provenance, S_x maps each semantically typed support channel to a value in the lifted domain

$$\mathcal{S} = \{\bot_\mu\} \cup [0,1], \qquad \bot_\mu \preceq s \;\text{ for every } s \in [0,1], \tag{2}$$

A_x records the scope over which each channel is declared applicable, and L_x is lineage. The lift matters for the propositions below. A channel can be *unavailable*, written $\bot_\mu$, and an unavailable channel must count as weaker than any numeric value it could have carried, otherwise "no emitted channel value is greater" would be silent about the case where a channel disappears entirely. Ordering $\bot_\mu$ below all of $[0,1]$ makes that case formal rather than merely intuitive, and $\preceq$ on $\mathcal{S}$ is the usual order on $[0,1]$ extended with $\bot_\mu$ as least element.

Provenance is a set of ancestry atoms drawn from $\{C, G\}$, for captured-derived and generated-derived sources. Mixed ancestry is $\{C, G\}$ and is produced only by union; it is never a primitive. The absence of verified evidence is a distinct state $\bot$. It is not an atom, and it is never silently mapped onto C or G. The presentation labels CAPTURED, GENERATED, MIXED and UNVERIFIED are rendering choices; the normative fields are P, S, A and L. The word REAL is not used as a provenance state anywhere in the model, the implementation or the serialisation.

4.2. The claim order and its meet

Write $\widehat{\mathcal{C}} = \{\bot\} \cup (2^{\{C,G\}} \setminus \emptyset)$ for the set of provenance claims, ordered by

$$q_1 \sqsubseteq q_2 \iff q_1 = \bot \;\text{ or }\; (q_2 \neq \bot \text{ and } q_1 \supseteq q_2), \tag{3}$$

read "q_1 is no stronger than q_2". A larger atom set is a weaker, less specific claim, and $\bot$, which asserts nothing, is the bottom element. The relation is reflexive, antisymmetric and transitive, so it is a partial order. Two claims always have a greatest lower bound but need not have any upper bound at all: $\{C\}$ and $\{G\}$ have none, since an upper bound would have to be a non-empty subset of both. The structure is therefore a meet-semilattice rather than a lattice, and its meet is

$$q_1 \sqcap q_2 = \begin{cases} \bot & \text{if } q_1 = \bot \text{ or } q_2 = \bot, \\ q_1 \cup q_2 & \text{otherwise.} \end{cases} \tag{4}$$

This is an instance of a standard construction: an idempotent, commutative, associative combination with an absorbing element, of the kind used to annotate derived data in provenance semirings [6] and to classify objects in information-flow lattices [7]. We record it explicitly rather than deriving it as a novelty.

4.3. The complete-source operator

Let a representation operator T produce output element y from the *complete required source set* $D_y = \{x_1, \ldots, x_k\}$. Two notions of dependency must be distinguished here, because they diverge for some operators. The *representational* dependency of y is the source content y stands for: what a listener at y's position is hearing a representation of. The *computational* dependency is the set of source samples that numerically influence y's value. The contract governs the representational reading: D_y is every source element whose content y represents. The kernel footprint then reconciles the two where signal processing makes them bleed together. A resampling filter makes neighbouring content numerically influence y, and since the direction of that influence cannot be distinguished from representation at the contract's level of abstraction, the footprint conservatively enlarges D_y to cover it. The one v1 operator for which the two readings diverge completely is amplitude normalisation, whose scalar gain is computed from the whole signal while each output sample represents exactly one source sample; Section 9 states the choice made there and its tested alternative. Write $D_y^\mu = \{x \in D_y : \mu \in A_x\}$ for the sources that declare channel μ applicable. The contract requires:

(i) Provenance $P_y = \bigsqcap_{x \in D_y} P_x$, that is, the exact union of the required source provenance sets, with $\perp$ absorbing and with no invented ancestry.

(ii) Support if $P_y \neq \perp$, $D_y^\mu \neq \emptyset$, $A_y(\mu) \neq \emptyset$ and every member of D_y^μ supplies a value, then $S_y(\mu) = \min_{x \in D_y^\mu} S_x(\mu)$; otherwise the channel is *unavailable*. The applicability condition belongs in this antecedent: without it a channel narrowed to an empty scope under (iii) would still be required to carry a value, which would be a number declared applicable nowhere.

(iii) Applicability the *canonical* scope is $A_y(\mu) = \bigcap_{x \in D_y^\mu} A_x(\mu)$, and an empty intersection makes the channel unavailable. A *conforming* operator may emit any $A'_y(\mu) \subseteq A_y(\mu)$, withholding the channel outside it, but never a broader one. Conformance is the containment, since a narrower scope cannot promote; the equality names the canonical reduction, which is what the reference implementation emits. Statements quantifying over operators that satisfy (i) to (v), Theorem 1 in particular, are to be read in this conforming sense, so that an operator emitting a narrower scope is covered rather than excluded by the equality.

(iv) No partial subsets a channel is available on the output only if every required source declares it applicable *and* supplies a value: formally $D_y^\mu = D_y$ is required, not merely $D_y^\mu \neq \emptyset$. A channel applicable to some required source but unreported by it is unavailable, and so is one that is not applicable to some required source at all. In neither case is it computed from the remaining sources. The two halves are the same principle, and the second is what makes Proposition 4 hold for channels: without it, enlarging D_y by a source that declares an otherwise inapplicable channel would raise that channel from unavailable to a value, which is a strengthening.

(iv′) Unverified absorption for channels if any required source is in state $\perp$, every numeric channel on the output is unavailable.

(v) Lineage $L_y = \bigcup_{x \in D_y} L_x$, identifying the complete required source set. The union alone does not place those sources in time, and is not asked to: the evidence record is carried per evidence-homogeneous interval, so L_y names the assets required by *that* interval and the interval's own sample bounds carry the placement. A source asset contributing several noncontiguous regions therefore appears in several interval records, each with its own bounds, rather than as one token a consumer would have to disambiguate.

Requirements (iv) and (iv′) are where this model is strictly stronger than its spatial predecessor, and both need a justification. The strengthening in (iv) is the demand that $D^{\mu}_y = D_y$ rather than merely $D^{\mu}_y \neq \emptyset$: a channel not applicable to some required source is withheld rather than computed from the rest. It costs expressiveness, since sources carrying different channel schemas now yield no channel at all on a shared interval rather than a partial one, and that is the price of Proposition 4 holding for channels at all. The strengthening in (iv′) is the companion rule for $\perp$. An element in state $\perp$ carries no evidence record at all, so nothing is declared about any channel over the span it occupies. Computing a numeric channel from the remaining, verified sources would be exactly the partial-subset computation that (iv) forbids, applied to the represented span rather than to the channel. Making every channel unavailable is the conservative resolution, and it is tested separately in the conformance suite.

4.4. Properties

Proposition 1 (Exact union; no promotion)**.** *Under (i), $P_y \sqsubseteq P_x$ for every $x \in D_y$, and no atom appears in P_y that appears in no required source. When no required source is in state $\perp$, P_y is exactly the set of atoms present on the required sources.*

Proof. Immediate from (4). If some required source is $\perp$ then $P_y = \perp$, and $\perp \sqsubseteq q$ for every q by (3), so the ordering clause holds and P_y contains no atom at all. Otherwise P_y is the union of the required source sets, which contains each P_x and is therefore $\sqsubseteq P_x$ by (3), and equals the set of atoms present on those sources. In both cases membership of an atom in P_y requires membership in some P_x, so no ancestry is invented. □

Proposition 2 (Unverified non-promotion)**.** *If any $x \in D_y$ has $P_x = \perp$ then $P_y = \perp$, and no numeric channel is emitted.*

Proof. Absorption in (4) gives $P_y = \perp$; requirement (iv′) then withholds every channel. □

Proposition 3 (Typed support and scope non-promotion)**.** *For every channel μ, $S_y(\mu) \preceq S_x(\mu)$ in $\mathcal{S}$ for every $x \in D^{\mu}_y$, and $A_y(\mu)$ is no broader than any $A_x(\mu)$.*

Proof. The minimum over a set is bounded above by each member; requirement (ii) refuses to emit unless every applicable source supplies a value, so the minimum is taken over the complete applicable set rather than a subset. Repeated intersection in (iii) cannot enlarge a set. □

Theorem 1 (Closure under composition)**.** *A finite composition of operators each satisfying (i)–(v), including (iv′), satisfies the same requirements, and the composed claim is no stronger than the meet over the union of all ancestral required source sets.*

Proof. Associativity and commutativity of $\sqcap$ make the composed provenance the meet over the union of the ancestral sets, so no ancestor is dropped or invented; absorption of $\perp$ is preserved inductively because no later operator receives a verified element in place of an unverified ancestor. For a channel μ, each operator either refuses the channel or emits a value bounded by its applicable inputs, and transitivity of $\preceq$ bounds the final value by every applicable ancestor. Repeated intersection cannot broaden a scope, and lineage composes by union of the per-operator source maps. □

The property in Theorem 1 is the standard consequence of combining in a meet-semilattice and is stated here for completeness rather than as a new result. The next proposition is specific to the temporal setting and is what makes the contract implementable against real processing software.

Proposition 4 (Footprint monotonicity)**.** *If $D_y \subseteq D'_y$ then the claim computed from D'_y is no stronger than the claim computed from D_y, and for every channel μ the value computed from D'_y is $\preceq$ the value computed from D_y in $\mathcal{S}$.*

Proof. Enlarging the index set of a union can only add atoms, and by (3) a superset is a weaker claim; adding a source in state $\perp$ sends the claim to $\perp$, which is weaker still.

For channels the argument runs through (iv), which requires $D^{\mu}_y = D_y$ for availability. Suppose μ is available on D'_y. Then every member of D'_y declares μ applicable and supplies a value, so every member of $D_y \subseteq D'_y$ does too, and μ is available on D_y as well; the emitted value is a minimum over the larger set D'_y, which is no larger than the minimum over D_y. Suppose instead μ is unavailable on D_y. Some member of D_y then fails to declare μ applicable, fails to supply a value, or is in state $\perp$; that same member lies in D'_y, so μ is unavailable on D'_y as well, and $\perp_\mu \preceq \perp_\mu$. The remaining case is an added source whose scope empties the intersection in (iii): requirements (ii) and (iii) then withhold the channel, giving $\perp_\mu$, which by (2) is $\preceq$ every element of $[0, 1]$. Every case therefore moves down $\mathcal{S}$.

The condition $D^{\mu}_y = D_y$ carries the weight here. Under the weaker condition $D^{\mu}_y \neq \emptyset$ the statement would be false, and concretely so: take $D_y = \{x_1\}$ with μ not applicable to x_1, so the channel is unavailable, and enlarge to $D'_y = \{x_1, x_2\}$ where x_2 declares μ applicable with some value $v \in [0, 1]$. The channel would rise from $\perp_\mu$ to v, and $\perp_\mu \prec v$ for every such v, so enlarging the required-source set would have strengthened the claim. $\square$

Proposition 4 is the practical result of this section. It means an implementation does not need the exact dependency set of a filtered operator: a *conservative over-approximation* is always safe, and only under-approximation can promote. That converts an intractable requirement, knowing exactly which input samples every output sample depends on inside a third-party codec, into a tractable one: declare a bound, and test that the bound holds.

Signal transparency (P8). Finally, the contract must not change the audio. We define the canonical essence of an asset as the SHA-256 of its decoded stream rendered to signed 16-bit little-endian PCM at the asset's own rate and channel count, and require that essence be identical whether the EM assertion or the baseline assertion is attached. Whole-file hashes are unusable for this purpose, because embedding a manifest changes the file by construction; the distinction matters and is measured in Section 7.5.

5. Audio instantiation

5.1. Temporal intervals

All positions are sample indices on a named asset and all intervals are half-open $[a, b)$, matching the C2PA convention that "All start times are inclusive of that moment in time, and all end times are, by default, exclusive of it" [3, §18.2.2.3]. A source timeline is a gapless ordered partition of an asset into evidence-homogeneous intervals.

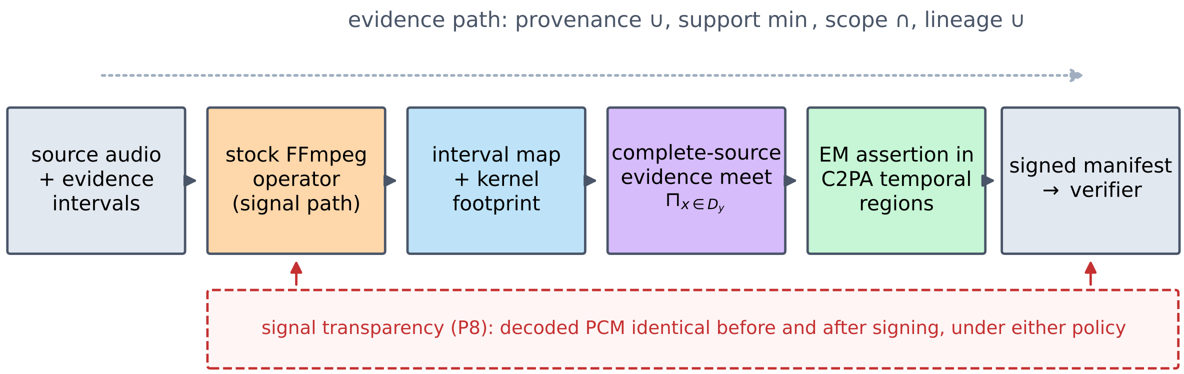

Figure 2: Where the contract sits. Stock FFmpeg performs the transformation; the interval map with its declared kernel footprint determines the complete required source set; the complete-source meet produces the emitted evidence; the result is serialised as C2PA temporal regions of interest and signed. The evidence path never touches the signal path, which is what signal transparency (P8) measures.

The v1 support channel. Version 1 carries one typed channel, `capture-support`, declared applicable over the single scope `digital-asset`. It is the degree to which an interval's own record supports a capture claim for that interval, on the lifted domain $\mathcal{S}$; it is not a detector output and is not calibrated against any acoustic measurement. What this instantiation demonstrates about the channel is therefore schema and conformance, that a typed channel aggregates by minimum, absorbs correctly, intersects its scope and survives the signed transport, and not that the particular numbers carry forensic meaning. A deployment supplying a calibrated score would inherit the algebra unchanged and would owe its own validation of what the score means. The corpus and transport experiments assign the two declaration-derived values 0.90 to captured-derived and 0.10 to generated-derived source intervals, so a mixed interval aggregates to 0.10 by the minimum in (ii); the exhaustive enumeration of Section 7.1 instead sweeps the domain positionally, so that every value in $\mathcal{S}$ is exercised rather than two. One channel is enough to exercise the value rules this instantiation relies on, since (ii) governs the value, (iv) the partial-subset case and (iv′) absorption, and all three are checked over the full enumeration and carried through the signed transport. It is deliberately not a strong test of (iii). A single scope cannot exhibit disjoint or partially overlapping scopes, nor a channel applicable to one required source and not another, and that last case is the one Proposition 4 turns on. Those cases are covered instead by a synthetic battery over 5 scopes, reported in Section 7.1.

An operator emits a set of *map pieces*. A piece states that output range $[o_a, o_b)$ represents source range $[s_a, s_b)$ on a named asset, together with a footprint radius in source samples. Where pieces overlap, as in a mix or an overlay, the required source set of an output interval is the union over every piece covering it, so an overlapped region represents both sources. The emitted output partition is the pull-back of every source evidence boundary across every piece, which gives each emitted interval a stable complete source set.

5.2. *The v1 operator set and its kernel footprints*

Table 3 gives the eight operators in scope with their mapping and declared footprint. Pitch shifting is deliberately not a ninth primitive: its common implementations are compositions of operators already in scope, a rate change reinterpreted plus a compensating time stretch, so its contract behaviour follows from Theorem 1 applied to those two, with the composed footprint the composition of theirs; a phase-vocoder implementation would instead declare its own analysis-window footprint, exactly as time stretching does. A named regression test exercises the composite. Denoising, source separation, speech enhancement and generative editing are out of scope: they are not representation-only, and a generative edit legitimately introduces new ancestry rather than propagating existing ancestry.

The footprint entries deserve comment, because a reviewer will look for an off-by-one. For a polyphase resampler the radius follows from the pinned filter length and the rate ratio. For MPEG-1 Layer III the long window is 1 152 samples with 50% overlap and the encoder introduces a further delay, giving a kernel radius of 1 728, to which one granule of frame-alignment guard, 576 samples, is added. That derivation is necessary but not sufficient, and we are explicit about why: the pinned encoder is `libmp3lame` with its bit reservoir enabled, which we now set explicitly rather than inherit, and a reservoir carries bits across frame boundaries, so the encoder is stateful in a way no window-and-delay argument fully captures. The declared 2 304 is therefore an *empirically validated bound for this frozen configuration*, confirmed by the containment probes of Section 7.6, which measured an actual reach of 1 555 source samples, rather than a codec-wide support theorem. A different bitrate, encoder or reservoir setting requires the bound to be re-probed. For overlap-add time stretching it covers the analysis window scaled by the tempo factor. For a frame-granular selector such as silence removal, the exact retained-sample model and the implementation's frame-aligned cut points differ, and the declared value covers that difference. In every case the declared value is a bound, not an estimate, and Proposition 4 makes an over-declared bound safe. Two separate experiments check these numbers against software we do not control, and it is worth being exact about which quantity each one bounds. Section 7.4 characterises temporal mapping deviation, by comparing predicted with actual output length, which is what informs the implementation margin. Section 7.6 tests whether the *declared* footprint contains the measured source influence. It does not independently validate the analytical column: for two operators the measured reach exceeds the analytical support and is contained only by the declaration as a whole, which is the honest reading of those measurements and the reason the declaration rather than either component is what an implementation uses.

5.3. The baseline reference policy

The comparison policy throughout is *boundary-only, primary-parent inheritance*: the derived output inherits from its parent ingredient, evaluated at the first and last source elements of the retained region, ignoring kernel footprints. It is a **constructed reference policy**. It is not a description of the behaviour of any shipping product, and no claim is made here about what any named tool does. Its motivation is the specification text: a derived asset's relationship to its input is `parentOf`, defined as "The current asset is a derived asset or asset rendition of this ingredient" [3, Table 10, §18.16.3], and the minimal way to inherit from a parent over a retained region is to read that region's endpoints. The policy is minimal by design because the point of the comparison is to isolate one mechanism, not to caricature an opponent.

The baseline combines two independent shortcuts, boundary-only inheritance and blindness to the kernel footprint, so the experiments also evaluate the two intermediate policies that remove one shortcut each, and report all four (Section 7.3).

Table 3: The v1 operator set. The analytical column is a support radius derived from the pinned algorithm configuration; the margin covers implementation behaviour that the analysis does not capture, such as frame-granular cut points and a correlation search range; the declared footprint is their sum, is what the implementation uses, and is the quantity Section 7.6 probes. For two operators the measured influence exceeds the analytical radius and is contained by the declaration, so the columns are two contributions to one bound rather than two independently validated bounds. All values are in source samples for the configurations used in the experiments. By Proposition 4 an over-declared bound can only weaken the emitted claim, while an under-declared bound is a conformance failure. The normalisation row is a representational footprint: its scalar gain depends computationally on the whole signal, and Section 9 states the choice and its tested alternative.

Operator	Source mapping	Analytical	Margin	Declared	Basis
trim / crop	1:1 on the retained range	0	0	0	exact
concatenate	1:1 per part	0	0	0	exact
resample 16 to 8 kHz	$n_{\text{out}} \cdot f_{\text{in}} / f_{\text{out}}$	33	64	97	polyphase FIR
transcode (MP3)	1:1	1 728	576	2 304	MDCT window + delay; see note
transcode (FLAC)	1:1	0	0	0	lossless
amplitude normalisation	1:1	0	0	0	scalar gain; see note
time stretch 1.10	$t_{\text{src}} = \tau\, t_{\text{out}}$	529	2 048	2 577	overlap-add window
silence removal	explicit retained runs	0	4 096	4 096	frame-granular selector
mix / overlay	1:1 from every covering source	0	0	0	sample-wise sum

5.4. Signed transport

Evidence is carried as a namespaced C2PA assertion, `mx.aurtech.emaudio.evidence`, with one entry per output evidence interval. Each entry gives a temporal region of interest as an `npt` range [3, §18.2.2.3], the sample-exact bounds alongside it because `npt` strings are lossy at sample resolution, the provenance atom list (`null` for unverified, never an atom), the available support channels, their applicability scopes, and the lineage set. Derivation records the source as an ingredient with relationship `parentOf`; the specification requires a corresponding `c2pa.opened` action carrying a hashed-URI reference to that ingredient assertion [3, §15.11.3.2], which the claim generator inserts. Composition records each source as a `componentOf` ingredient carrying its full source manifest, with a `c2pa.placed` action referencing each component's ingredient assertion [3, §18.16.3], so the transport exercises exactly the composition vocabulary whose missing aggregation rule motivates the paper (Section 7.7). The full schema, with a worked manifest, is Supplementary Note S4.

The transport is a bridge, not a re-implementation: signing and validation are performed by the official `c2patool`, so the transport guarantees are the tool's. The choice of C2PA is not load-bearing for any claim in this paper. The contract is defined over source sets and typed evidence, and the same assertion can be carried in a detached signed sidecar; the experiments use C2PA because it is the deployed standard and because doing so tests the mapping against a real validator.

6. Methods

6.1. Study design

The experiments separate three questions that are easy to conflate: whether the implementation realises the stated guarantees (a mathematical question, answered by exhaustive enumeration and by differential testing); whether a competing metadata rule produces counterexamples on adversarial and on realistic input (a mechanism question, answered by frozen fixtures and a labelled corpus); and what the contract costs (an engineering question, answered by timed repetitions). Exact algebraic properties and finite-state classifications are reported as deterministic counts and are not assigned p-values. Fixture failure frequencies characterise the frozen fixture family and are a **mechanism stress rate**: they are not, and must not be read as, an estimate of how often any deployed system promotes provenance.

6.2. Deterministic fixtures

Experiment A enumerates every source word of length 1 to 8 over $\{C, G, \perp\}$, giving 9 840 words, and pushes each through up to ten operator configurations. Two configurations require a minimum word length: the inner trim needs at least three elements and silence removal at least two. So the 3 single-element words receive eight configurations, the 9 two-element words nine, and the remaining 9 828 words the full ten, giving 98 385 cases. Each case is checked against the full property suite and against a separate closed-form oracle that recomputes the expected state directly from the operator's nominal source ranges without using the evidence algebra; for zero-footprint operators the states must match exactly, and for kernel operators the emitted claim may only be weaker.

Experiment B freezes 10 000 timelines of 64 source intervals each, generated from a fixed seed, in which the majority of intervals are captured and one to four *non-boundary* intervals are replaced by generated or unverified evidence. Each is pushed through composition chains of depth 1 to 5 and, separately, through each operator alone. Because the anomalies are deliberately placed away from the boundaries, the baseline's failure rate on this family is high by construction; to make that explicit the experiment includes a control arm in which anomaly positions are drawn uniformly over all positions, for which the baseline's promotion probability has the closed form $\binom{n-2}{k}/\binom{n}{k}$ for k anomalies of a single type over n positions. Agreement between the measured rate and that closed form is an analytic check on the harness.

6.3. Mixed-origin audio corpus

600 clips are built deterministically, each of the form captured | generated | captured with sample-exact ground truth. Captured segments are crops of LibriSpeech `dev-clean` [17], licensed CC BY 4.0 with the licence text read from the file shipped inside the archive; generated segments are produced locally by eSpeak NG formant synthesis from phrases written for this study. No speaker is identified, no voice is cloned, and no speaker-identity claim is made. Segment durations are matched so that the generated region is a modest interior interval rather than a synthetic clip with captured fragments attached. Licences, versions, URLs, access dates and checksums are recorded in `DATA_LICENSES.md` in the repository.

Because the evidence algebra reads no acoustic feature, the result cannot in principle depend on signal character; a supplementary robustness arm demonstrates this empirically rather than leaving it to argument, rebuilding 50 clips with a modern neural TTS (Piper, a VITS-architecture engine, with a voice trained on the public-domain LJSpeech corpus) in place of formant synthesis, and overlaying 50 clips with locally generated pink noise at -18 dB so that every ancestry boundary is acoustically buried.

Concatenation and every subsequent transformation are performed by **stock FFmpeg through its command line, with no project code in the signal path**. This matters for the same reason a third-party generalizer mattered in the spatial setting: the artefact that either does or does not exhibit promotion is produced by software the author does not control, and only the evidence policies are under study.

6.4. Metrics

We report promotion counts and rates, lineage-omission counts, unverified-to- verified promotions, support promotions, conformance failures, decoded-essence mismatches, manifest validation outcomes, model-versus-implementation sample deviations, runtime and metadata size. We deliberately do not report accuracy, precision, recall, F_1 or AUC: no classifier is evaluated, and detector metrics would invite the wrong comparison. Timing is reported as medians with interquartile ranges over 30 repetitions on a single machine: an arm64 system running macOS 26.5.2, Python 3.11.15, FFmpeg 9.0.1 and `c2patool` 0.27.2. The exact processor, operating-system build and every tool version are recorded in `results/PREFLIGHT.txt`, which is regenerated on each run.

6.5. Reproducibility and AI use

Every number in this manuscript is emitted by code into `results/machine_readable/`, converted into a LaTeX macro by a generator script, and used in the text through that macro; no result is typed by hand. Every figure is rendered from the same files. `run_all.sh` exits non-zero if any conformance check fails. Generative-AI tools were used as programming and review assistants during development of the verification tooling and in preparing the manuscript; all reported outcomes were produced by executing the committed deterministic code, and the author reviewed and verified every claim, calculation, code-derived number and reference.

7. Results

7.1. Exhaustive conformance

Over 9 840 source words and 98 385 operator cases, the property suite and the closed-form oracle together evaluated 885 828 checks with 0 failures, including 363 explicit composition cases. Every named regression test passes: 30 of 30, with 0 failures, and both counts appear as their own row in Table 7 rather than being folded into the comparison columns, which do not apply to them.

That enumeration carries one channel over one scope, which is what the audio instantiation uses, so it cannot reach the applicability cases of (iii). A separate synthetic battery covers those: over 5 scopes, chosen so that pairs are disjoint, nested and partially overlapping, it enumerates 441 source-set enlargements and checks each against Proposition 4, with 0 violations.

The battery earns its place, because it is what the audio corpus could not have told us. An earlier version of requirement (iv) withheld a channel only when a required source declared it applicable and reported no value, leaving a source to which the channel was not applicable at all to be silently skipped. Under that rule the enumeration of this section still passed every one of its 885 828 checks, because in the v1 instantiation every source that is not $\perp$ declares the one channel and every $\perp$ source is absorbed by (iv′). The battery fails it in 80 of its 441 cases, each an enlargement that raised the channel from unavailable to a value. The monotonicity check had the same blind spot in the same place: it compared a channel's value only when the channel was present on both sides, so the one transition that breaks the proposition was the one comparison it skipped. Both are fixed, and the battery runs in the pipeline.

The suite has teeth. During development it failed on the overlay operator, where the implementation evaluated overlapping map pieces independently and so emitted a captured claim and a generated claim over the same output interval instead of one mixed claim. That was a genuine defect in our own code, found by the test named for it, and fixed by making the output partition global across pieces. We report it because a conformance suite that has never failed is evidence of nothing.

7.2. *Adversarial timelines*

On 10 000 frozen timelines, boundary-only inheritance promoted the provenance claim on 9 455 timelines (94.5%) at composition depth 1 and on 9 455 at depth 5; complete-source propagation produced 0 promotions summed over every depth, 0 unverified-to-verified promotions, 0 lineage omissions and no support promotion. Boundary-only inheritance omitted required lineage on 10 000 timelines at depth 1 and reported 7 222 unverified spans as verified.

The rate does not rise with depth, and that is the finding rather than a flat line on a chart: the loss is established at the first derivation and is irreversible, because once the interior evidence has been discarded no downstream operator can recover it (Figure 3, left). Across the 9 operator configurations applied singly, baseline promotion ranged from 94.3% to 100.0%, and complete-source propagation produced 0 promotions in total (Figure 3, centre).

In the control arm, where anomaly positions are uniform over all positions rather than confined to the interior, measured baseline rates agreed with the closed-form prediction to within 0.0070 in absolute rate (0.70 percentage points) in every cell (Figure 3, right). This is the honest reading of the adversarial arm: its near-unity rate is a property of the deliberate construction, arithmetically predictable, and carries no information about deployed software. The informative quantity is the exact zero on the other side.

7.3. *Which shortcut matters: a policy ablation*

The baseline combines two shortcuts, and it is fair to ask which one does the damage. Removing them one at a time over 10 000 timelines per arm gives an unambiguous answer (Table 4). In the *interior* arm the anomaly lies inside the nominal represented range, so only the inheritance axis can see it: both boundary-only policies promoted on every timeline (10 000 and 10 000) and both complete-source policies on none. In the *footprint* arm the anomaly lies just outside the nominal range but inside the operator's declared kernel footprint, so only the footprint axis can see it: both footprint-blind policies promoted on every timeline (10 000 and 10 000) and both footprint-aware policies on none.

Neither axis is redundant and neither is sufficient. Complete-source aggregation without a declared footprint fails exactly the case that filtered operators create, and a declared footprint without complete-source aggregation fails exactly the case that interior edits create. Only the full contract clears both arms, which is the argument for specifying them together rather than separately.

7.4. *Mixed-origin audio through stock FFmpeg*

On 600 clips, complete-source propagation recovered the evidence interval structure exactly, every boundary at sample resolution, for 600 clips, and identified the generated interval with its exact bounds for 600 clips, with no detector, no training and no acoustic feature of any kind. The ground truth here is constructed rather than discovered, and that is the point: the evidence originates at the capture and generation steps, and the question this experiment answers is whether it *survives* a chain of representation-only transformations, not whether it could be inferred from the signal. Nothing in this experiment supports a claim that generated audio can be identified without evidence.

Figure 3: Boundary-only versus complete-source inheritance on 10 000 deterministic timelines. Left: promotion does not grow with composition depth because the loss is established at the first derivation and is irreversible. Centre: per-operator rates when each operator is applied alone; complete-source is zero for every operator and is drawn as a marker at the origin, since bars of zero extent would be invisible. Right: control arm with anomaly positions uniform over all positions, measured against the closed form $\binom{n-2}{k}/\binom{n}{k}$; note the truncated vertical axis (80–100%), chosen so that any divergence between measurement and prediction is visible rather than compressed. Percentages quantify mechanism stress on a frozen fixture family; they are not real-world prevalence.

The robustness arm behaves identically: 50 of 50 neural-TTS clips and 50 of 50 noise-buried clips recovered their full interval structure exactly, with 0 and 0 promotions respectively. Under the noise overlay the captured segments correctly become MIXED (captured content plus a generated noise source) while the generated segment correctly stays GENERATED, since union adds no new atom. Boundary-only inheritance promoted the whole-asset claim to captured-only on 600 clips and omitted required lineage on 600; the complete-source policy produced 0 promotions and 0 lineage omissions.

Across 8 transformations and 4 800 transformation runs (Figure 4; Table 7), boundary-only inheritance promoted on 3 600 runs and complete-source propagation on 0, with 0 lineage omissions. Two transformations produced *no* baseline promotion, and we report that as readily as the rest: overlay declares the generated source as a separate top-level ingredient, so a boundary-only reading of two ingredients still sees both, and silence removal happened to retain runs whose boundaries fell inside the generated region. Not every operator promotes, and a paper claiming otherwise would be wrong.

The interval model was validated against FFmpeg's actual output on every run where the output was PCM and on a 100-clip subset for the coded containers. The largest absolute deviation between predicted and actual output sample count was 681 samples for silence removal, 471 for time stretching, 31 for MP3 and 0 for resampling. For all transformations the declared mapping margin covered the observed deviation.

That measurement bounds one thing and not another, and the distinction matters enough to state plainly. Agreement between predicted and actual output length validates the *mapping and alignment* of the interval model. It does not establish the *support containment* that Proposition 4

Table 4: Policy ablation over 10 000 timelines per arm. The interior arm places the anomaly inside the nominal represented range; the footprint arm places it outside that range but inside the declared kernel footprint. Each shortcut fails exactly one arm, and only the full contract clears both.

Policy	Inheritance	Kernel footprint	Interior anomaly	Footprint anomaly
B0	boundary-only	no	10 000 (100%)	10 000 (100%)
B1	boundary-only	yes	10 000 (100%)	0 (0%)
B2	complete-source	no	0 (0%)	10 000 (100%)
B3	complete-source	yes	0 (0%)	0 (0%)

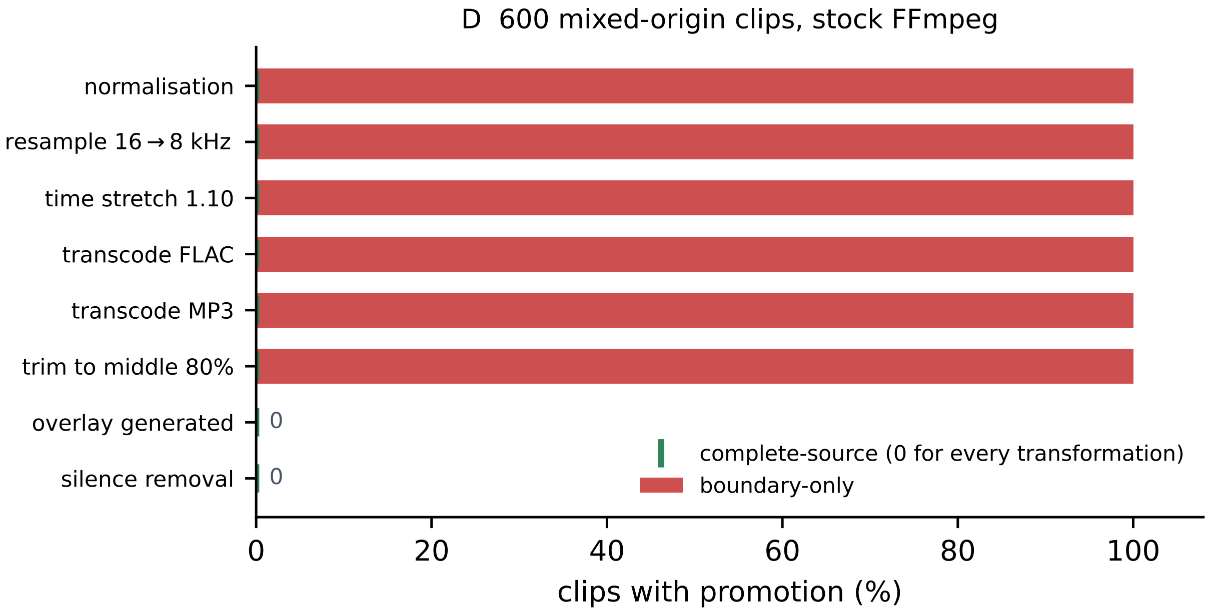

Figure 4: Promotion by transformation over 600 mixed-origin clips processed entirely by stock FFmpeg. Complete-source propagation is zero for every transformation, so its series is drawn as a marker at the origin rather than as bars with no extent. Overlay and silence removal produce no baseline promotion either, for the structural reasons given in Section 7.4.

actually requires, because an operator can emit exactly the predicted number of samples while depending on source samples outside the declared footprint. Containment is tested separately and directly in Section 7.6. Re-running each transformation and comparing decoded essence gave 0 mismatches, so the processing path is deterministic on this machine.

Two parts of this experiment could have failed, and one did. The zero promotion count under complete-source propagation follows from Proposition 1 once the required source set is correct, so on its own it checks that the implementation matches the algebra; on overlay, during development, it did not. The genuinely falsifiable measurements are the other two: whether the interval model predicts the output length of software we do not control, and whether the declared mapping margin bounds that deviation. Support containment is a third falsifiable measurement and is made separately in Section 7.6. Either could have come out wrong, and the declared footprints would then have had to be re-declared and the build would have failed.

7.5. *Signed transport, derivation and provenance loss*

Across 3 containers and 120 signed round-trips (Table 8), signing succeeded and validation reported the trusted state under the declared local anchor for 120 assets, with 0 validation failures. The EM assertion returned byte-equivalent from the validator for 120 assets. After derivation with stock FFmpeg, re-signing with the parent as an ingredient, and re-validation, 120 derived assets validated and 120 recorded the `parentOf` relationship.

Signal transparency holds exactly. Decoded essence was identical before signing, after signing with the complete-source assertion, and after signing with the baseline assertion, with 0 mismatches against the pre-signing essence and 0 between the two policies. Over the same assets the whole-file hash changed in 120 of 120 cases, which is why the comparison is defined on decoded PCM: a file-level test would have reported a change in every case and told us nothing. Reproducing the pipeline from a clean clone makes the same point from the other direction. Comparing the two runs asset by asset, every decoded-essence hash agreed and no signed-file hash did, because each signature carries its own timestamp. A reproducibility check defined on file bytes would have declared this pipeline irreproducible; defined on decoded essence, it reproduces exactly.

Under the five provenance-loss conditions over 60 clips (Table 9), a removed manifest yielded UNVERIFIED in 60 cases, an asset modified after signing yielded INVALID in 60, and re-encoding without manifest preservation yielded UNVERIFIED in 60. No condition other than a valid manifest ever yielded CAPTURED, and there were 0 violations of the classification requirement. We do not claim that stripping is detected as malicious intent; the result is that the classification degrades in the conservative direction.

7.6. *Kernel-support containment*

Proposition 4 needs the declared required-source set to contain the actual dependency set, and that is a statement about which source samples influence which output samples, not about how many output samples there are. We test it directly. The probe covers 7 of the eight transformations in the matrix; overlay is excluded because a single-source temporal impulse is not the relevant test for it. Its required-source set is by construction every source covering the output interval, and the sample-wise sum has no temporal kernel at all, so there is no footprint to under-declare. For each configuration, each of 4 signal contexts and each of ten probe positions, two 16 384-sample sources are built that are identical except at one sample, where one carries an added impulse. Both are pushed through the same stock FFmpeg command and both outputs are decoded. The entire difference between the paired outputs is attributable to that one source sample, because the inputs differ nowhere else; every other source sample still contributes to both outputs, but identically. Containment holds when the probe position lies inside the declared required-source range of every output sample whose value it changed. The comparison is made on 16-bit PCM at a threshold of one least significant bit, so an influence far below audibility still counts as a dependency.

Two properties of the probe had to be got right, and getting them wrong first is what showed why they matter. The carrier must not be silence: against silence every operator here appears to map one source sample to one output sample, because an impulse in an empty buffer is simply copied through. And the perturbation must be proportionate to the carrier. An earlier version used a fixed impulse amplitude, which was roughly two hundred times the carrier in the low-level context; that does not perturb an adaptive operator's input so much as replace it, and both the lossy encoder and the overlap-add stretcher responded by re-deciding globally rather than locally. The perturbation is now a fixed multiple of each context's peak. Against real content at a proportionate level, the same operators spread a single sample widely.

The 4 contexts are chosen so that the operators are probed where each behaves differently rather than at one operating point: a steady low-frequency tone, the ordinary case; the same tone at roughly one part in a hundred amplitude, where a lossy encoder allocates fewest bits and quantisation is coarsest; repeated sharp onsets with exponential decay, which drives an overlap-add stretcher hardest; and a sum of six inharmonic partials, spectrally crowded so that the encoder's masking decisions are most active. All four are generated arithmetically with no random source; the exact parameters are in `experiments/support_containment.py`. Lossy transcoding spreads a single source sample across up to 1 823 output samples and does so at every probe position, and resampling across up to 33. Overlap-add time stretching is the interesting case: its spread is strongly position-dependent, reaching 2 023 output samples at the 7 probe positions that land inside an overlap region and one sample at the rest. A declaration derived from the median would have been badly wrong for that operator, which is why the containment test is run against the worst case rather than a typical one.

Across 280 probes, 52 005 output samples were influenced and 0 of them fell outside the declared support (Table 5). Every context produced influence for every operator, so no row passes vacuously by the probe simply having no effect.

That zero was not free, and the route to it is the argument for running this experiment at all. A shorter probe than the one reported here passes trivially for the large-footprint operators, because a declared footprint of 2 577 samples spans a whole short source; lengthening the probe until the footprint covers a fraction of it made the test able to fail, and it did. Silence removal failed: a frame-granular selector emitted a frame beginning 3 277 source samples before a retained run's start, well beyond the margin declared for it at the time. That is an under-declaration, which is the one direction Proposition 4 does not forgive. The band was raised to 4 096 samples, above the measured reach, and the operator now contains with a margin of 819. We report this rather than quietly shipping the corrected number because it is the clearest evidence that the declared bounds are measured rather than asserted. The bands are also not merely adequate but reasonably tight, which matters because a larger one would be equally safe and more dilutive. Resampling's influence reached 32 source samples against a declaration of 97, MP3's reached 1 555 against 2 304, and time stretching's reached 1 906 against 2 577. Each declaration sits above its measured reach with headroom rather than far above it, which is what justifies these particular numbers instead of merely permitting them.

7.7. C2PA-native heterogeneous composition

The strongest form of the standards argument is executable, and we execute it. For 30 fixtures, a captured source and a generated source are each signed with their own `digitalSourceType`; stock FFmpeg concatenates them; the composition is signed with both sources as `componentOf` ingredients carrying their full source manifests, a `c2pa.placed` action referencing each component's ingredient assertion, and the EM assertion computed by the complete-source rule over the two component timelines. The composition is then derived across the ancestry boundary and re-signed with `parentOf`.

The action vocabulary deserves a note, because the specification offers two candidates. `c2pa.mixed` is defined for the case where "multiple, previously placed (via `c2pa.placed` actions) audio ingredients (e.g., stems, vocals, drums, bass, etc) are combined and optionally transformed" [3, Table 8]. Our fixture concatenates its two sources in sequence rather than summing them, and we interpret sequential concatenation as placement rather than mixing. We label that a *documented profile decision* rather than a normative reading, because the specification does not settle the boundary

Table 5: Kernel-support containment. Each operator is probed with a proportionate single-sample impulse at ten positions in each of 4 signal contexts. The measured reach is the furthest any influence extended beyond the nominal, footprint-free source range, so containment compares two concrete quantities rather than resting on an assertion: it holds exactly when the reach does not exceed the declaration. Headroom is the difference, so a small headroom means a tight declaration and a large one a conservative one. † Amplitude normalisation's declared footprint is *representational*: an output sample represents one source sample, while the scalar gain depends computationally on the whole signal. The probe therefore confirms the representational reading and does not show that the computational dependency is zero, which it is not.

Operator	Declared	Measured reach	Influenced samples	Headroom	Outside
amplitude normalisation†	0	0	40	0	0
resample 16 to 8 kHz	97	32	1 131	65	0
silence removal	4 096	3 277	28	819	0
time stretch 1.10	2 577	1 906	10 971	671	0
transcode to FLAC	0	0	40	0	0
transcode to MP3	2 304	1 555	39 763	749	0
trim to the middle 80%	0	0	32	0	0

and we have no validator consensus to cite for it. The specification does not settle the question either way: it neither states that concatenation is not mixing, nor excludes `c2pa.remixed`, which covers "transformation of a previously recorded audio asset" including re-arranging [3, Table 8]. What is not a matter of interpretation is that `c2pa.mixed` presupposes `c2pa.placed`, so the placed actions are required under any reading, and an overlay composition in which two sources sum over a shared interval would warrant `c2pa.mixed` in addition. The choice does not affect the contract: the aggregation rule reads the ingredient graph and the interval map, not the action label.

All 30 compositions validated as trusted under the declared anchor, with both `componentOf` ingredients recorded in 30 cases and both `c2pa.placed` references resolved in 30; the EM assertion returned byte-equivalent in 30 cases; the whole-asset aggregate was MIXED with the captured and generated intervals individually recoverable in 30; and the derived asset stayed MIXED and trusted in 30 of 30 validated derivations, with 0 essence mismatches. This is the precise object the vocabulary-versus-reduction distinction describes: the component histories are all present in the signed graph, and a consumer walking the ingredients finds both source manifests and both source types. The aggregate interval claim exists only because the EM assertion computes it. Nothing in the validation pipeline required the aggregate to be conservative; the contract supplied that property, inside the standard's own extension points. A representative composition manifest is frozen as a fixture in the repository.

7.8. The cost of conservatism: claim dilution

Zero promotion is only half of a useful result, because safety alone is cheap: declaring the whole asset as every sample's dependency is perfectly safe under Proposition 4 and perfectly useless, since every interval near an ancestry change would collapse to the weaker claim. We therefore also measure what the declared footprints cost. The *dilution fraction* of a derived output is the fraction of its samples whose footprint-aware evidence record is strictly weaker (a weaker provenance claim, a lost channel, or a lower support value) than the record computed over the nominal represented content, which on this corpus is exact by construction.

Measuring this immediately caught a second defect in our own implementation, and we report it for the same reason as the first. The reference implementation originally applied footprint widening at the granularity of the emitted interval partition: an interval whose widened source range touched a neighbouring ancestry was assigned the weakened claim over its *entire* extent, so a single resampling of a corpus clip reported a dilution fraction of one: safe, by footprint monotonicity, and uselessly conservative. The fix refines the output partition at the pull-backs of each source boundary shifted by $\pm$ the footprint, after which dilution is confined to footprint-wide bands at ancestry boundaries. The two-language differential had not caught this because its fixtures used source intervals narrower than every declared footprint, a geometry in which the coarse and the exact treatments coincide; the oracle corpus now includes a wide-interval battery in which they do not, and the fixed implementation agrees with the per-sample oracle on all 10 860 cases.

After the fix (Table 6), 4 of 8 transformations dilute nothing at all. Among them is overlay, whose nominal required-source set already contains every covering source, so the footprint adds nothing. The remainder dilute the footprint-wide neighbourhood of each ancestry boundary: median dilution is 0.86% of output samples for resampling, 20.24% for MP3 transcoding, 31.34% for silence removal and 22.64% for time stretching, with a single-transformation worst case of 58.33%. These fractions must be read against the corpus geometry: each band has a *fixed radius in samples* (the declared footprint of Table 3, which already includes the guard), and the clips average 2.85 s, so the same bands on a one-hour asset would occupy a fraction three orders of magnitude smaller. The fraction shrinks with asset duration; the absolute band does not.

Composition is where conservatism genuinely bites, and we report it as measured rather than as hoped. The chain is, in order: trim to the middle 80%, resample 16 to 8 kHz, transcode to MP3, amplitude normalisation, and time stretch by a factor of 1.10 at the post-resampling rate, deliberately front-loading a rate change so that the large MP3 and time-stretch bands are scaled by it. Along this chain the median dilution reaches 85.74% by depth 5, with a worst case of 99.99%. The per-depth series shows each operator contributing its own bands and nothing else: depth 4 is identical to depth 3 to the last digit, because the operator added there is amplitude normalisation, whose declared footprint is zero. On a short asset a chain of large-footprint operators therefore approaches total dilution. Whether a rate change also rescales the bands inherited from upstream follows from the interval model but is not isolated by this design, since the chain's only rate change occurs while no upstream bands yet exist; we state it as a property of the model rather than as a measurement. Asset length is the other variable, and it is worth separating from chain composition rather than reading both out of one number. Holding the chain fixed at a routine three-operator edit (trim to the middle 80%, normalise, transcode to MP3) over an asset with one interior generated segment, and varying only duration, dilution is 25.13% at the corpus median clip length, 2.40% at 30 seconds and 0.240% at five minutes (Table 6). Each tenfold increase in duration divides the fraction by ten. That is what fixed-width bands predict when the number of ancestry boundaries is held fixed, which it is here, and it is what the measurement shows. It is not a claim that the relation holds for an arbitrary chain or an arbitrary boundary count, since more boundaries would place more bands. This is the real price of declaring conservative bounds instead of knowing exact codec dependencies, and it is a cost a deployment can buy back without losing safety: any tighter dependency map, such as an encoder that reports its true delay or a codec-specific footprint smaller than our pinned bound, reduces dilution and, by Proposition 4, can never introduce promotion, provided it remains an over-approximation of the truth. Measuring the cost is what makes that engineering trade visible; a contract that only reported its zero promotion count would have hidden it.

Table 6: Claim dilution: the fraction of output samples whose footprint-aware evidence is strictly weaker than the evidence over the nominal represented content, per transformation over 600 clips and along a composition chain over 200 clips. This is the price of the declared footprints, which combine analytical support with implementation margin, when the nominal map is exact.

Transformation	Clips	Median dilution	Max dilution	Clips affected
amplitude normalisation	600	0.00%	0.00%	0
overlay a generated source	600	0.00%	0.00%	0
resample 16 to 8 kHz	600	0.86%	1.39%	600
silence removal	600	31.34%	58.33%	600
time stretch 1.10	600	22.64%	44.92%	600
transcode to FLAC	600	0.00%	0.00%	0
transcode to MP3	600	20.24%	32.56%	600
trim to the middle 80%	600	0.00%	0.00%	0
composition depth 1	200	0.00%	0.00%	—
composition depth 2	200	1.06%	1.59%	—
composition depth 3	200	50.87%	75.84%	—
composition depth 4	200	50.87%	75.84%	—
composition depth 5	200	85.74%	99.99%	—
same depth-3 chain, 2.85 s asset	1	25.13%	—	—
same depth-3 chain, 30 s asset	1	2.40%	—	—
same depth-3 chain, 300 s asset	1	0.24%	—	—

7.9. Two-language differential

The Python reference implementation works with interval algebra over pulled-back source boundaries. The Node.js oracle evaluates evidence per output sample by brute force and run-length-encodes the result. The frozen corpus spans two interval geometries: one in which kernel footprints are wider than every source interval, and one in which they are far narrower, the regime that distinguishes a per-sample treatment of footprint widening from an interval-granularity one, added after the dilution experiment exposed exactly that difference (Section 7.8). On 10 860 cases the two implementations agreed on interval structure, provenance state, lineage set and channel availability with 0 disagreements, and the maximum absolute difference in any emitted support value was 0. Both implementations are author-written, and the oracle was deliberately built with no interval algebra at all, and evaluates evidence numerically per output sample, so a defect in the reference implementation's pull-back arithmetic cannot recur in the oracle as the same defect class. The same author can still make correlated conceptual errors, which is why this is a differential test against transcription and interval-arithmetic error, and not evidence of independence from the author (Section 9).

7.10. Cost

Complete-source bookkeeping cost a median 0.93 ms per audio-minute against 0.28 ms for boundary-only, a ratio of 3.36× computed from the unrounded medians rather than from the two figures as printed, which differ in the last place they are quoted to, and which is 0.156% of the FFmpeg time it accompanies (Table 10; Figure 5). Signing took a median 9.7 ms per asset (IQR 9.5–9.9) and validation 8.2 ms (IQR 7.8–8.5).

Manifest overhead was a median 14 854 B per asset, of which the EM assertion was 1 268 B. The manifest cost is dominated by a fixed per-asset component (certificate chain, COSE structure,

Table 7: Main results. Deterministic fixture rates characterise the frozen fixture family and are a mechanism stress rate, not a population prevalence estimate.

Check	Cases	Boundary-only	Complete-source
A Exhaustive finite-state conformance			
Source words over $\{C, G, \perp\}$, length ≤ 8	9,840	—	—
Operator cases	98,385	—	—
Contract checks	885,828 checks, 0 failed		
Named regression tests	30 tests, 30 passed, 0 failed		
B Deterministic adversarial timelines (10 000 frozen fixtures, 64 intervals)			
Chain depth 1	10,000	9,455 (94.5%)	0
Chain depth 3	10,000	9,455 (94.5%)	0
Chain depth 5	10,000	9,455 (94.5%)	0
Lineage omissions (depth 1)	10,000	10,000	0
Unverified → verified (depth 1)	—	7,222	0
C–D Mixed-origin audio corpus, stock-FFmpeg processing			
Exact propagation of constructed interval evidence	600	—	600
amplitude normalisation	600	600 (100.0%)	0
overlay a generated source	600	0 (0.0%)	0
resample 16 to 8 kHz	600	600 (100.0%)	0
silence removal	600	0 (0.0%)	0
time stretch 1.10	600	600 (100.0%)	0
transcode to FLAC	600	600 (100.0%)	0
transcode to MP3	600	600 (100.0%)	0
trim to the middle 80%	600	600 (100.0%)	0
H Two-language differential oracle			
Frozen cases	10,860	—	0 disagreements

hard binding), so we report it per asset and give the variable part separately: the assertion grows linearly in emitted evidence intervals at 333 B and 14.0 μs per interval, measured up to 256 intervals. Evidence cost therefore scales with how many times ancestry actually changes along the timeline, not with duration, which is the property a long-form or streaming deployment needs. These are single-machine medians for a Python reference implementation and are not language-independent constants.

One further caveat is worth stating precisely, because the within-run interquartile ranges above do not capture it. Repeating the whole pipeline on this one machine moved the absolute cost per audio-minute by tens of percent between runs, while the ratio to the boundary-only baseline held to within a few percent. The reason is structural rather than incidental: both arms are timed in the same process on the same machine, so whatever sets the absolute scale on a given run, thermal state, frequency scaling or competing load, moves the two arms together and cancels in their quotient. The ratio is therefore the quantity to carry across machines, and the absolute figures should be read as order-of-magnitude evidence that the bookkeeping is cheap rather than as reproducible constants. A reader reproducing these numbers should expect the ratio to survive and the milliseconds not to.

Table 8: Signed transport. Columns are, in order: assets signed; assets whose validation state was trusted under the declared local anchor (the test credential is not on the C2PA Conformance Program trust list); derived assets that validated after re-signing; derived assets recording the `parentOf` ingredient relationship; assets whose EM assertion returned byte-equivalent; and decoded-essence mismatches before versus after signing.

Container	Assets	Trusted	Derived	`parentOf`	Assertion	Essence
FLAC	40	40	40	40	40	0
MP3	40	40	40	40	40	0
WAV	40	40	40	40	40	0

Table 9: Provenance-loss behaviour. The requirement is that the reported state degrade conservatively; no claim is made that a missing manifest indicates intent.

Condition	Clips	Reported state
valid manifest	60	MIXED (60)
manifest removed	60	UNVERIFIED (60)
asset modified after signing	60	INVALID (60)
re-encoded without manifest preservation	60	UNVERIFIED (60)
valid derived manifest with complete lineage	60	MIXED (60)

8. Discussion

What is established. That a complete-source rule is implementable over media time against real processing software; that it eliminates promotion on every fixture and every corpus clip tested, where a minimal boundary-only policy does not; that it can be carried in a deployed signed transport without altering the audio; and that its cost is a fraction of a percent of the processing it accompanies.

What is not. No claim is made that any deployed system promotes provenance. The baseline is a constructed reference policy and the fixture rates are a mechanism stress rate. Nor is the algebra new: the non-amplification property is an instance of a meet in a semilattice, familiar from provenance semirings [6] and information-flow models [7]. Readers who already accept that algebra should read the contribution as the temporal instantiation, the kernel-footprint treatment, the transport and the measurements, which is where the work actually was.

Relation to the provenance standard. C2PA authenticates signed assertions about asset history, and its vocabulary already preserves what this problem needs: component ingredients with their own manifests, generative source types, and temporal regions. The layer addressed here is the reduction over that vocabulary: whether the evidence assertion emitted by a derived representation is conservative with respect to every source interval that representation actually contains. C2PA can carry the component histories; the contract defines the aggregate claim a conforming operator must compute from them. This work complements the standard rather than criticising it, and it is implemented entirely inside the standard's own extension points, a namespaced assertion and temporal regions of interest, without requiring any specification change. Where the specification's security properties are discussed, we cite a publicly available security analysis, a preprint at the time of writing, rather than asserting anything ourselves [11].

Relation to detectors and watermarks. These are complementary and answer different questions. A detector examines a signal; a watermark marks content at generation time; the contract here

Table 10: Cost, as medians over 30 repetitions on one machine. These are measurements of a Python reference implementation, not language-independent constants.

Quantity	Median	IQR
EM bookkeeping	0.926 ms / audio-minute	0.919–0.933 ms / audio-minute
Boundary-only bookkeeping	0.276 ms / audio-minute	0.275–0.278 ms / audio-minute
EM / boundary-only ratio	3.36×	—
EM as a fraction of FFmpeg time	0.156%	—
C2PA signing	9.7 ms / asset	9.5–9.9
C2PA validation	8.2 ms / asset	7.8–8.5
Manifest size overhead	14,854 B / asset	14,854–14,855
EM assertion	1,268 B / asset	1,268–1,269
EM assertion marginal cost	333 B / evidence interval	—

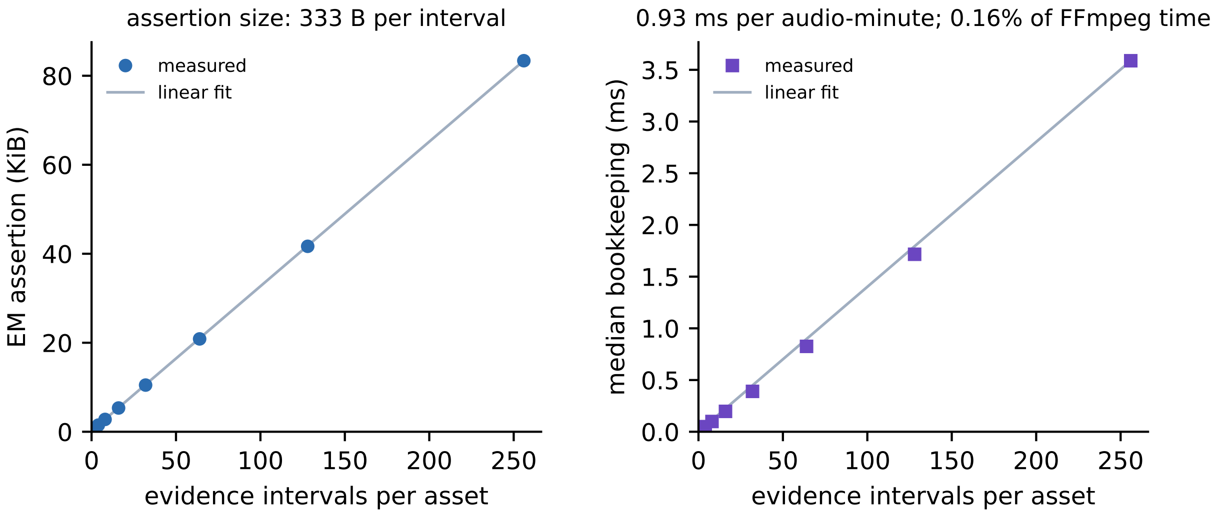

Figure 5: Cost of carrying interval evidence. The emitted assertion and the bookkeeping time both grow linearly in the number of evidence intervals, not in audio duration, so cost scales with how often ancestry changes along the timeline.

constrains what a derived asset may assert given what its sources asserted. A pipeline can run all three. The contract is at its most useful precisely where detectors are weakest, on short interior edits inside otherwise genuine recordings, and it is useless where detectors are essential, namely media that arrive with no evidence at all.

Attack surface. The contract narrows one surface and leaves others untouched. It removes the possibility that a conforming chain silently strengthens a claim, and it removes the incentive to launder a mixed-origin asset by passing it through a benign transformation. It does nothing against a malicious signer, a compromised key, or an actor who strips provenance entirely and starts a new, dishonest chain. Cryptographic validity proves that an assertion was signed, not that it is true.

Deployment path. The contract deploys as a library alongside the processing step, not as a pre- or post-processor: the operator invocation already knows its own parameters, which determine the interval map and footprint, so the natural integration point is the same code that assembles

Table 11: Claim–evidence boundary. Rows deliberately separate contract conformance, mechanism demonstration, transport behaviour and cost, so that evidence from one row is not used to authorise a claim in another.

Claim class	Evidence used here	Not established by that evidence
Operator conformance	Propositions 1–4 and Theorem 1; exhaustive finite-state enumeration; two-language differential	That any deployed system conforms; the prevalence of non-conformance
Mechanism on constructed fixtures	10 000 frozen adversarial timelines with a closed-form control arm	Any population rate; the behaviour of any named product
Mechanism on realistic audio	600 labelled mixed-origin clips processed by stock FFmpeg	Balanced coverage of speakers, languages, codecs or editing styles
Interval-model fidelity	Predicted versus actual output sample counts on every PCM run and on a 100-clip subset for the coded containers, against declared margins	Kernel-support containment, which is a different property, measured separately; and that the bounds hold for other codecs, builds or configurations
Kernel-support containment	280 probes: ten positions in each of 4 signal contexts across 7 operator configurations, compared on decoded PCM at one least significant bit	Containment for probe positions, signals, codecs or configurations not tested; the probe bounds the operators as configured here, not the operator class
Signed transport	120 round-trips through the official tool across 3 containers	Conformance-program trust; security of the C2PA cryptographic layer
Signal transparency	Decoded-PCM SHA-256 identity before and after signing under both policies	Perceptual equivalence claims; bit-exactness across different decoders
Cost	30 timed repetitions on one declared machine	Cost in a compiled implementation or at broadcast scale
Conservatism cost	Per-sample dilution against constructed exact ground truth	Dilution against the true (unknowable) dependencies of a third-party codec

the FFmpeg command line or codec call. The editor computes the map, the library computes the aggregate, and the existing signing step carries the assertion. Nothing observes or modifies the signal path, which is why signal transparency holds by construction rather than by care. Because the assertion schema and the temporal regions are ordinary C2PA extension points, adoption requires no specification change; a claim generator that emits the assertion and a consumer that reads it can interoperate through a conforming C2PA validation pipeline, which we demonstrate with the official `c2patool` and have not tested against other validator implementations. The schema is published with the repository as a baseline for derived-audio manifests.

An information-flow reading. The contract can equally be read as an integrity property in the classical information-flow sense [7, 8]: evidence labels form a meet-semilattice with an absorbing bottom, and the requirement that a derived interval's label be the meet over everything that flowed into it is precisely a "no write up" discipline for evidential authority. Readers from the security community may find that framing the shortest proof that the construction is sound; the temporal contribution is the mapping from operators to flows via interval maps and footprints.

Composing with adjacent work. The contract addresses aggregation and says nothing about surviving transport, which is what chunk-localised perceptual binding addresses [14] and this work does not.

The two are complementary rather than competing: a pipeline could use perceptual chunk binding to recover evidence after a manifest is stripped, and the contract here to decide what the recovered evidence entitles a derived interval to claim. We have not built that combination.

Extension. Nothing in Section 4 refers to audio. The instantiation cost for another medium is the interval algebra and the footprint table: frames and regions for video, spatial regions for images, character ranges for text. We have not built those, and we do not claim them.

9. Limitations

Independence. The second-language oracle and the evidence policies are author-written. The differential is a test against transcription and interval-arithmetic error, not evidence of author independence. Two components are outside our control and were deliberately chosen for that reason: all audio processing is stock FFmpeg through its command line, and all signing and validation is the official `c2patool`. The pipeline has been reproduced from a clean clone of the tagged release, which is a weaker check than it sounds and is worth stating precisely: it establishes that the code and reproducibility workflow are self-contained conditional on the two checksum-pinned external artefacts, and that no result depends on a file left in a working directory; it was performed by the author on the same machine, so it is not independent reproduction. A third-party reimplementation of the contract, and a reproduction by someone else on different hardware, would strengthen external validity further; neither exists yet.

Corpus. 600 clips from one read-speech corpus and one formant synthesiser, at 16 000 Hz, is a deliberate mechanism corpus, not a balanced design over speakers, languages, recording conditions, codecs or editing styles. It supports the claim that the mechanism is real and that the rule works on realistic audio; it does not support any rate claim about audio in general.

Footprints are declarations. The kernel footprints are conservative bounds declared for the exact processing configurations used here. They are validated against those configurations and no others. A different FFmpeg build, resampler setting, codec or tempo factor requires the bound to be re-declared and re-tested; Proposition 4 guarantees that erring large is safe, not that our particular numbers transfer. The re-declaration need not be manual, but it needs two passes rather than one, because the analytical and implementation-margin components of Table 3 bound different things. A *mapping-guard* calibration pushes a deterministic reference signal through the unknown configuration and sets the mapping margin from the maximum output-length deviation plus a margin, which is the measurement the transformation matrix already makes. A *support-footprint* calibration sweeps a single-sample impulse across the input and sets the kernel radius from the widest measured influence plus a margin, which is the measurement Section 7.6 makes. Substituting the first for the second is exactly the conflation this paper had to correct in its own draft: output length can be exact while the dependency is wider than declared. Kernel radii could equally be derived from codec specifications where those state window and delay constants. Both passes are implemented as `tools/calibrate_footprint.py`, which reports the measured reach, the largest output-length deviation and a recommended declaration for each operator, and exits non-zero if any declaration fails to contain its own measured reach. Its support pass is a probe at ten prespecified positions in each of 4 contexts, not an exhaustive sweep, so it cannot discover a mode transition occurring only at an untested position; the positions are the operators' own cut points and edges, where an under-declaration shows up first. It retains every per-probe record rather than the maximum alone, so a reader can see which position and context produced the reach. A self-test shrinks a known declaration below its own measured reach and fails unless the run rejects it, which is what distinguishes a tool that certifies declarations from one that has been shown to detect a bad one; it runs in the pipeline. Run against the v1 operators it reproduces the reaches of Section 7.6 exactly, which is the cross-check that the tool measures the same quantity the validation does. It separates the requirement from the policy: containment of the measured reach is what soundness needs, whereas the recommended margin, a fraction of the reach with a floor, is a conservatism choice the tool reports but does not enforce. Two of the declarations shipped here sit below that advisory margin while containing their reach, and are marked as such rather than quietly raised. Both passes inherit their safety from footprint monotonicity, since a calibration that errs large cannot promote.

Measured footprints are local to the operating points tested. The containment probe measures how far a proportionate local change propagates at the signal contexts and probe positions used. Adaptive operators do not have one dependency structure: a lossy encoder re-allocates bits and an overlap-add stretcher re-selects its correlation lag when the input changes enough to alter those decisions, and we saw exactly that when an early version of the probe perturbed a low-level carrier with an impulse two hundred times its amplitude, after which one source sample influenced most of the output. The correct conclusion is narrow. The declared footprints characterise dependency around the tested operating points; they do not establish a universal support radius over every input-induced mode transition, and an operator remains representation-only whether or not its internal decisions are input-adaptive. Note also what is *not* being claimed: a whole-asset footprint would bound any such transition trivially, so the open question is the size of a local radius, not whether a bound exists. Where an implementation cannot supply or validate a local radius, declaring whole-asset dependency remains available and remains safe under Proposition 4, at the dilution cost quantified in Section 7.8.

Normalisation is a modelling choice. Amplitude normalisation is the operator where representational and computational dependency (Section 4) diverge completely: the scalar gain is computed from the whole signal, but the content an output sample represents is one source sample. The contract governs representational dependency, so the default footprint is zero; the maximally conservative alternative, in which every output sample represents the whole signal, is implemented and tested as well. Both readings are defensible. The reported results use the representational reading, and a deployment that prefers the computational one pays the corresponding dilution cost and gains nothing in promotion safety, since neither reading can promote.

Trust anchor. The signing credential is locally generated and the trust anchor locally configured. A trusted validation state in these results means trusted under that anchor, not conformance-program trust.

Replay and the analogue hole. A microphone recording a loudspeaker that is playing synthetic speech produces audio that is genuinely captured at the sensor layer while its semantic source is synthetic. Every capture-provenance system has this limitation and this one is no exception: the contract would propagate a captured claim, correctly at the layer it models and misleadingly at the layer a reader may care about. The honest response is layering: capture mechanism, media-generation ancestry and speaker identity as separate evidence channels. Version 1 deliberately stays at the digital-asset and source-lineage level. Nothing here detects re-recording.

Speaker identity and truth. Identity is not provenance: knowing an interval was captured says nothing about who spoke. Neither is truth: a captured recording can contain a lie, and the contract says nothing whatsoever about semantic content.

10. Conclusion

A representation-only transformation may change, simplify, compress, aggregate or reorganise content, but it must not silently increase the evidential authority of the output relative to the complete source material that output represents. In audio that principle has an exact form: provenance is the union over the complete required source set, same-channel support is the minimum over it, applicability is intersected, lineage is the union, and an unverified source interval is absorbing. The underlying algebra is not new, and we have said so; what this work provides is the executable temporal contract, the kernel-footprint treatment that makes it implementable against real codecs, a signed transport inside an existing standard, and measurements showing that the rule eliminates promotion on every fixture and clip tested, leaves the decoded audio untouched, and costs a fraction of a percent of the processing it accompanies. It does not detect deepfakes, it does not establish that speech is truthful, and a valid signature proves only that an assertion was signed, not that it is true.

Data availability

The mixed-origin corpus is generated deterministically by committed code from LibriSpeech `dev-clean` [17], licensed CC BY 4.0, fetched by a checksum-pinned script, and from locally generated eSpeak NG output. The robustness arm also uses a Piper neural text-to-speech voice trained on the public-domain LJ Speech corpus, fetched and checksum-verified by `tools/fetch_voice.sh` rather than committed, and pink noise generated locally by FFmpeg with a per-clip seed. No human-subject, proprietary or login-walled data is used, and no voice of any identified living person is cloned. Licences, tool and model versions, URLs, access dates, file digests, the noise-generation command and redistribution clauses are all recorded in `DATA_LICENSES.md`.

Code availability

The reference implementation, the second-language oracle, every experiment, every frozen fixture and every machine-readable result are released under the MIT licence, with pinned tool versions and one-command reproduction. The repository URL is supplied with the submission and printed here at proof stage; we state it that way rather than asserting public availability the reader cannot yet act on. The evaluated state is release tag v1.0.0. Its exact commit hash is recorded in `results/PREFLIGHT.txt`, which is regenerated on every run and shipped with the release; that report also carries every tool version and every number quoted in this manuscript, so a reader can check any figure in the text against the file the code wrote. A Zenodo archive with version and concept DOIs is deposited on acceptance and the DOIs are added at proof stage.

CRediT authorship contribution statement

A. Urias: Conceptualization, Methodology, Software, Formal analysis, Validation, Investigation, Visualization, Writing – original draft, Writing – review and editing.

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Declaration of competing interest

A. Urias is employed by and holds equity in Aurtech. No product of Aurtech is evaluated, named or compared against in this work, and the baseline is a constructed reference policy rather than a description of any product. A non-conformance found in the author's own implementation during development is reported in Section 7.1.

Declaration of generative AI and AI-assisted technologies

During preparation of this work the author used generative-AI tools to assist with development and review of the verification and reproducibility tooling, manuscript organisation, and language refinement. All reported outcomes were produced by executing the committed deterministic code rather than generated by these tools. The author reviewed and verified the scientific claims, calculations, code-derived results, references and final text, edited the material as needed, and takes full responsibility for the content of the publication.

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